



# Towards the Objective Characterisation of Major Depressive Disorder Using Speech Data from a 12-week Observational Study with Daily Measurements

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## Abstract

Our analysis focuses on identifying relations between properties of the voice and depression symptom severity. On a novel corpus of 3,374 longitudinal speech recordings from 71 patients clinically diagnosed with major depressive disorder (MDD), we use a statistical modelling approach to identify associations between depression symptom severity and 38 acoustic and cognitive features. Significant negative associations with daily within-individual fluctuations of depression include speaking rate and articulation rate. Furthermore, when analysing how the changes in speech-derived features covary over time with the change in depression severity, we find that the standard deviation of the pitch has a significant negative association, as well as the speaking and articulation rate. We also discover that several performance metrics derived from the cognitive tasks (digit-span and Stroop) have significant associations with fluctuations or changes in depression symptom severity.

**Index Terms:** longitudinal speech analysis, health, depression

## 1. Introduction

Depression is a serious medical condition and a leading cause of disability worldwide, with one in three women and one in five men experiencing major depression by the age of 65 [1]. Conventional methods of depression assessment and treatment are limited by the availability of trained clinicians [2]. This strain on services is exacerbated when the goal is to monitor depression longitudinally – i.e., not just assessing individuals to make an initial diagnosis and treatment recommendation, but also monitoring them frequently over time as they trial different treatments [3]. As such, designing systems that use consumer technologies such as smartphones to scale care is an active area of research, both in the clinical and direct-to-consumer spaces [4]. A critical component of these systems is the ability to perform the assessment of depression symptom severity. To match the evidence used by a clinician, such an assessment module should incorporate both the user's description of their well-being and life experience, as well as details on how they present (e.g., their tone of voice). Therefore, identifying *intra-individual* (and *inter-individual*) associations between depression and speech is an important sub-problem to enable such digital mental health applications. These biomarkers might also inform the study of depression more generally – e.g., identifying new features to be validated further in clinical trials [5].

Prior work on the relationship between speech and depression has sought to link the severity of depression (or its constituent symptoms) to impairment in different components of the speech production system [6, 7]. Interpretable features are designed to measure key properties of these components. For example, *phonation* or *voice quality* features such as the

*harmonics-to-noise ratio* (HNR) relate to the generation of the speech signal in the larynx, *spectral* features like the location and bandwidth of formants relate to the shaping of the vocal tract to modify the raw speech signal into phonemes, and *fluency* and *prosody* features quantify the timing of syllables and variations in pitch and loudness throughout the utterance.

In a 6-week longitudinal study of 35 patients with MDD, Mundt et al. studied the relationship between *fluency* and *prosody* features [8] where speech samples were collected over the telephone on a weekly basis. They find that the rate of speech production decreases with increased depression, with patients with higher levels of depression articulating fewer syllables per second and pausing for longer. Using the same data, Quatieri et al. computed *phonation* features of jitter, shimmer and HNR on extracted vowels from transcripts and observed that shimmer (positive association) and HNR (negative association) were significantly associated with overall depression severity [9]. These associations were also present when considering *psychomotor retardation* – a core symptom of depression related to impairment in the motor coordination system [9].

In an 18-month longitudinal study with 585 participants [10], Cummins et al. compute a battery of features that probe multiple components of the speech production system on speech samples collected by smartphone every two weeks and test their association with depression symptom severity. They found that fluency features such as speaking rate and articulation rate, as well as how loudly a participant speaks, are most reliably associated with depression symptom severity, all exhibiting a negative association. There have also been several studies on the cross-sectional association between acoustic features and depression [6, 7] as well as numerous machine learning analyses, e.g., using the *Distress Analysis Interview Corpus/Wizard-of-Oz* (DAIC-WOZ) dataset [11].

It has often been posited that intra-individual changes in speech properties are highly coupled with changes in depression [6, 7]. However, due to the paucity of datasets with frequent longitudinal measurements (e.g., daily) over a long observation window (e.g., several months), this coupling has not yet been tested at a fine-grained temporal resolution. To begin investigating this, we use a statistical framework to identify longitudinal associations between speech features and major depressive disorder. We ask two questions: first, are daily fluctuations in speech properties associated with daily fluctuations in depression? Second, does the change in a property of an individual's speech over time covary with their change in depression symptom severity? Our dataset contains very high-frequency measurements of speech and depression – both were requested daily from participants over a 12-week window. Thus, this dataset and analysis are novel as they allow testing for associations between speech and depression in intensive longitudinal data.

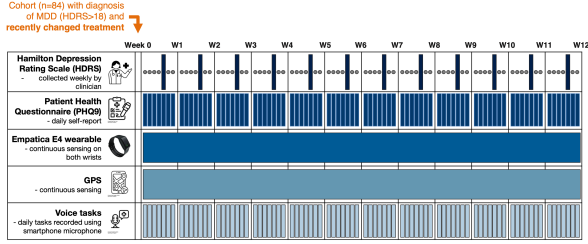


Figure 1: Participants provide survey, speech and physiological data daily throughout the 12-weeks.

## 2. Methodology

### 2.1. Study Protocol

We analyse daily speech recordings from a 12-week observational study of patients with major depressive disorder (MDD). The inclusion criteria for the study are: (i) that the patient should have MDD diagnosed with a total score of  $>18$  on the 17-item Hamilton Depression Rating Scale (HDRS) [12]; and (ii) that the patient should have recently changed their depression treatment (a step that was taken to increase the variability of their depression severity over the observation window). This study collected daily speech recordings and responses to the self-reported nine-item Patient Health Questionnaire (PHQ9) depression rating scale [13]. In addition, weekly HDRS scores are collected by clinicians assessing the patients, and wearable physiology and smartphone-based location data are also collected. The full study protocol is in Figure 1.

In this analysis, we focus on the speech and PHQ9 measures. The daily speech recordings were collected from a battery of speech tasks presented in a smartphone app. These tasks, in order, are: (i) hold a sustained vowel (“ahhhhh”); (ii) a diadochokinetic task repeating the syllables “/pa/ta/ka/” (PTK); (iii) three sentence reading tasks (in which the sentences to read are randomised across study days); (iv) two *digit-span* memory tasks where the participants are shown a 3- or 5-digit number and then a sentence and then are asked to recall the sentence followed by the number; and (v) a *Stroop test* with a low and high cognitive load condition where participants are asked to say the word written on the screen where the colour of the word is congruent or incongruent in the low and high cognitive load conditions, respectively. This results in 9 recordings per session. The daily speech tasks were completed after a reminder at 9am local time for the participants, while the daily PHQ9 survey was completed after a reminder at 3pm local time. The duration of recordings for each task is in Figure 2.

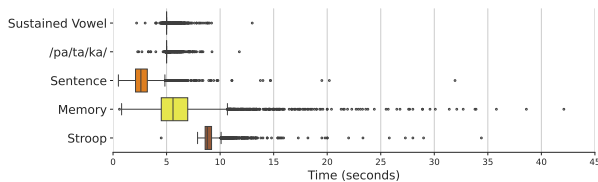


Figure 2: Recording duration by task type. There are 29,713 individual tasks recorded across 3,374 sessions (after exclusions). We see that individual tasks are generally short – median  $<10s$  for all task types, though there are some outliers of longer duration. The median length of the entire session of 9 tasks is 98s.

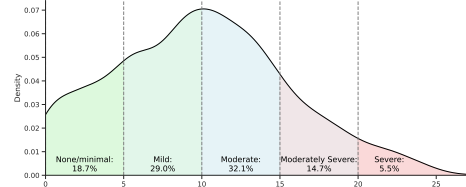


Figure 3: Distribution of total PHQ9 score for depression severity over all participants. We see that there is substantial density in all severity buckets. The intra-class correlation (ICC) is 0.60 suggesting a high degree of clustering within individual.

### 2.2. Dataset

84 participants were onboarded into the study, providing 5,046 PHQ9 scores and 3,940 audio sessions. The data exclusion criteria were: (1) there should be at least 10 observations per participant, so that we can appropriately test for *intra-individual* associations; (2) a session should not last longer than 60 minutes; (3) a recording must contain some speech per a voice activity detection (VAD) algorithm [16]; and (4) for tasks where we expect participants to say something intelligible (e.g., the prompted sentence), the transcripts should contain some language. We use *Nvidia NeMo* to transcribe [17].

71 participants provided enough clean data for this speech analysis – reducing the dataset to 3,374 clean audio sessions with an associated PHQ9 survey. The distribution of the PHQ9 scores across all participants is displayed in Figure 3, indicating a broad range of depression severity levels. The demographics of the cohort are: biological sex 84.5% female and 15.5% male; 12.7% asian, 11.3% black, 8.5% more than one race, 66.2% white, and the remaining race not reported; 12.7% hispanic; the mean (std) age is 30.8 (12) years with min age of 19 and max age of 70 years; 95.8% use English as their primary language.

We compute 38 features from the daily speech recordings using *Parselmouth* [18]. Our feature design is inspired by recent work [10, 19]. These features are categorised as follows

- **Fluency:** including speaking rate, articulation rate, phonation ratio, pause rate, and mean pause duration. In addition, for the PTK task, we compute the articulation rate separately and also the standard deviation in the intersyllable interval
- **Prosodic:** including loudness of the voice (intensity), as well as mean and variation in the pitch
- **Phonation:** relating to the production of the speech signal in the larynx, these quantify *voice quality*, and include the harmonics-to-noise (HNR) ratio, the spectral slope and tilt, and the cepstral peak prominence (CPP) [20]
- **Spectral:** relating to the filtering of a speech signal via vocal tract movements. We compute the mean and standard deviation in the frequencies and bandwidths of the first and second formants, as well as several moments of the spectra from 0.025s windows of the speech signal with stride of 0.005s
- **Cognitive:** we compute several features from the cognitive tests to quantify cognitive functioning. For the sentence reading tasks, we simply compute the *word accuracy* (WAcc) between the transcripts and the sentence prompt. For the memory tasks, we compute a WAcc separately for the sentence and the numbers. In addition, we compute various pause-related metrics. For the *Stroop* tasks, we compute all features as delta between the high and low cognitive load conditions. We compute a delta WAcc, several pause-related delta metrics, and the delta in the count of words and non-colour words

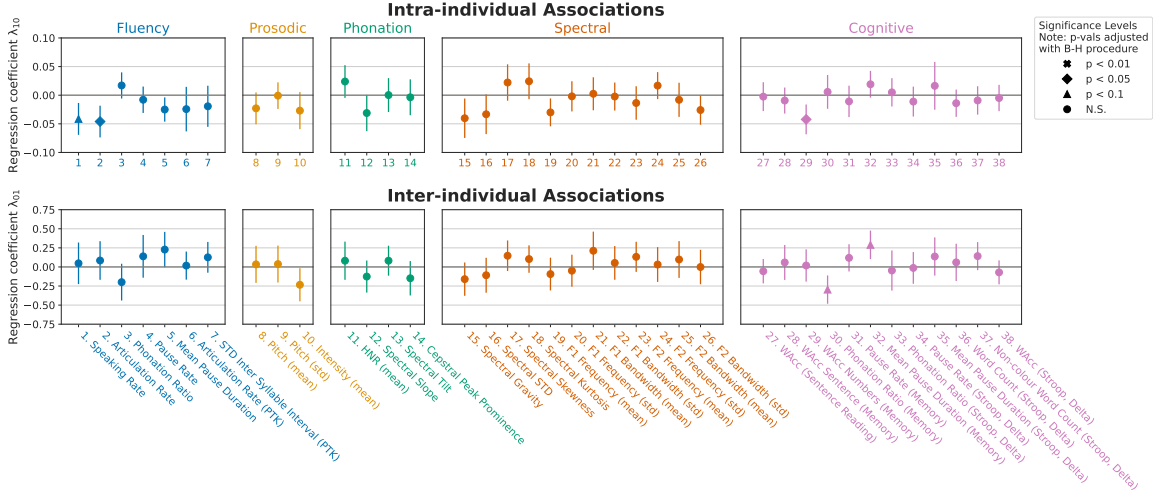


Figure 4: *Analysis 1. Linear mixed effects models reveal several features associated with depression severity. Positive coefficients indicate the feature increases with increasing depression. The confidence intervals (CI) are computed using cluster-robust standard errors. Unadjusted significance at  $\alpha < 0.05$  can be inferred where CI does not cross zero. False discovery rate (FDR) controlled significance using the Benjamini-Hochberg (B-H) procedure [14] is marked by shape. Coefficients are pseudo-standardised [15].*

Given we have multiple tasks per session, we average the features per task to get a single value of each feature for a given session. Not all tasks are appropriate for calculating each feature, so we map different tasks to different features. All tasks are used for *prosodic*, *phonation* and *spectral* features. *Fluency* features are only computed on sentence reading tasks, with two additional features computed from the PTK task. The cognitive features are derived from the corresponding cognitive tasks.

### 2.3. Statistical Modelling

**Analysis 1: Linear Mixed Effects (LME) Model:** This analysis is for our first research question: are speech properties associated with depression severity, either between individuals or within individuals (i.e., fluctuations relative to their overall baseline levels for the feature)? To this end, we use linear mixed effect modelling (LME) to test each feature in isolation. LMEs are used to analyse data with repeated observations. With  $y_{ti}$  as the PHQ9 total score and  $z_{ti}$  the speech feature for patient  $i$  at time  $t$ , we define the *level-1* equations for our LME model as:

$$y_{ti} = \beta_{0i} + \beta_{1i}z_{ti} + \beta_{2i}t + r_{ti} \quad (1)$$

$$\hat{z}_{ti} = z_{ti} - \bar{z}_i \quad (2)$$

$$\bar{z}_i = \mu(z_{ti}) \quad (3)$$

$$\text{Cor}(r_{ti}, r_{t'i}) = \rho_{|t-t'|} \quad (4)$$

Note we disaggregate within-individual from between-individual variability by centring each *time-varying* feature measurement by the individual mean,  $\bar{z}_i$ , to produce  $\hat{z}_{ti}$  [22]. Furthermore, we include time in the model to negate possible net time effects. This allows us to make more precise inferences about how speech production (and cognition) relate to depression severity. We include a first-order autoregressive structure in (4) for autocorrelation. Our model *level-2* equations are

$$\beta_{0i} = \lambda_{00} + \lambda_{01}\bar{z}_i + u_{0i} \quad (5)$$

$$\beta_{1i} = \lambda_{10} \quad (6)$$

$$\beta_{2i} = \lambda_{20} + u_{2i} \quad (7)$$

Here,  $\lambda_{00}$  is the intercept fixed effect,  $\lambda_{10}$  is the within-person fixed effect for the speech feature,  $\lambda_{20}$  is the fixed effect

for time. The between-person fixed effect for the speech feature is  $\lambda_{01}$ . We allow random effects  $u_{0i}$  and  $u_{2i}$  on the intercept and slope with time, respectively, but not on the slope with the time-varying speech feature  $\hat{z}_{ti}$  as we found this led to convergence issues. We focus on reporting  $\lambda_{10}$  and  $\lambda_{01}$  as these are the core effects of interest.

We compute cluster-robust standard errors of type *CRI* [23] to account for heteroscedasticity. We use the R libraries *nlme* for LME with *AR(1)* correlation structure and *clubSandwich* for cluster-robust error correction [24, 25]. We standardise coefficients using the *pseudo* method where within-individual variance is used for  $\lambda_{10}$  and within-group variance for  $\lambda_{01}$  [15].

**Analysis 2: Intra-Individual Change Over Time:** Our second analysis relates to our second research question: how do *changes over time* in features covary with *changes over time* in depression severity? To study this, we performed a two-step procedure. First, we fit simple linear regressions individually (i.e., one per participant) for the PHQ9 score and time and the acoustic feature and time. We then extract the slope coefficients from these models and perform a linear regression between them in a second step. Formally:

$$y_{ti} = \psi_{0i}^y + \psi_{1i}^y t + \epsilon_{ti}^y \quad (8)$$

$$z_{ti} = \psi_{0i}^z + \psi_{1i}^z t + \epsilon_{ti}^z \quad (9)$$

$$\psi_{1i}^y = \phi_0 + \phi_1 \psi_{1i}^z + \delta_i \quad (10)$$

Here  $\psi$  are the coefficients of the regressions fit per individual  $i$  in the first step of the analysis, and  $\phi$  are the coefficients fit across individuals in the second step. We focus on reporting  $\phi_1$ , which determines if there is a relationship between the change in the value of a feature over time and the change in the value of patient depression. We report the results of a simple and robust linear regression (using *robustbase* [21] for the latter).

## 3. Results and Discussion

**Analysis 1:** Focusing on relations that are significant after a Benjamini-Hochberg adjustment for false discovery rate (FDR) [14], we observed that fluctuations in two *fluency* features – the *Speaking Rate* and *Articulation Rate* – are associated with fluctuations in depression severity relative to individual baselines

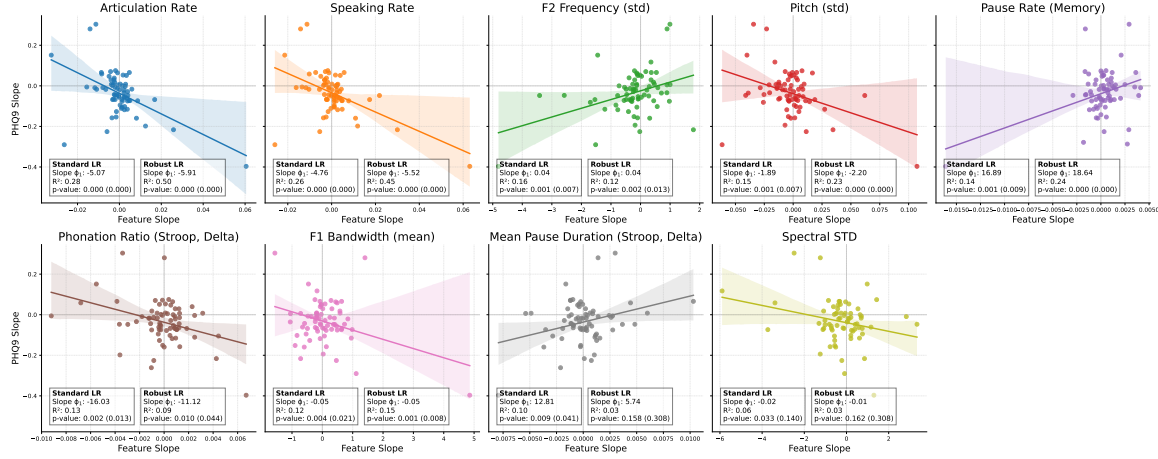


Figure 5: Analysis 2. Linear regression of PHQ9 time slope on speech feature time slope. We report the linear slope coefficient  $\phi_1$  (cf. Equation 10), sorted by  $R^2$  and filtered to only the features with significant p-values ( $\alpha < 0.05$ ). To control for outliers, a robust regression (Robust LR) using robustbase is also reported [21]. Benjamini-Hochberg corrected p-values are reported in parentheses.

(see Figure 4). This suggests that on days where participants are less depressed relative to their baselines, they speak faster. Importantly, this aligns with the findings of a recent longitudinal study [10]. For cognitive features, we also report a significant negative intra-individual association between the accuracy of recalling the number in the *digit-span* task and depression severity – on days where a patient is less depressed they recall the digits more accurately. We focus on intra-individual associations given the longitudinal nature of our study but still report inter-individual results in Figure 4 for reference.

**Analysis 2:** When considering the regressions between the individual-specific slope parameters of PHQ9 against time and each acoustic feature against time, *Speaking Rate* and *Articulation Rate* were again significantly associated features under this test (Figure 5). Here, this finding suggests a negative association between the direction of change in these features over time and the direction of change in depression symptom severity; in other words, when depression symptom severity decreases over time, speaking and articulation rates tend to increase (and vice versa). Interestingly, another acoustic feature that is significant in this analysis was the standard deviation in the pitch (i.e., how monotonously someone is speaking). We see that the change in this value over time is negatively associated with the change in depression severity: as someone becomes less depressed, their voice becomes less monotonous. While this feature is often not identified using approaches like Analysis 1 (e.g., in [10]), a similar change-based analysis did identify this as a potential biomarker [26]. We also note that several formant features, as well as pause-related metrics in the memory and Stroop tasks, were identified through this analysis. All results remain significant in the standard linear regression after controlling the FDR with the exception of *Spectral STD*. One important caveat to our findings is the presence of outliers in Figure 5. However, these associations appear to still be significant after using a robust linear regression to control for outliers.

**Discussion.** Our results show strong agreement with contemporary literature [10, 7]. We find that *Speaking Rate* and *Articulation Rate* are negatively associated with depression severity in two ways: daily fluctuations relative to baseline (Analysis 1) and when changing over time (Analysis 2). When combined with results in [10, 27], there is strong evidence that these are promising vocal biomarkers for depression. Interestingly, we

did not see any significant association between *phonation* (or *voice quality*) features such as HNR, as previous work identified these features to be significantly associated with depression [9]. An explanation for this absence could be the real-world nature of our smartphone recordings: not only might they possess more background noise relative to recordings collected in lab settings, but they are also subject to device-specific audio-filtering before being written to file [28]. Thus, important properties of the signal may have been removed upstream of our analysis.

We have identified three main limitations of our study. First, as discussed, the real-world nature of our study and use of smartphones can result in variable quality of recordings and the loss of valuable signals. Second, only modelling associations with linear models is a simplification when growth over time and relationships between features and depression could be nonlinear. Third, our speech task protocol was limited to short scripted speech tasks. We cannot probe linguistic features, and some acoustic features found not significant in this analysis may indeed be significantly associated with depression when measured from longer speech samples or different elicitation prompts (e.g., pausing and hesitating in free-response speech).

## 4. Conclusion

Utilising a novel longitudinal corpus of 3,374 recordings from 71 participants with clinically diagnosed major depressive disorder, we studied the longitudinal associations between interpretable acoustic and cognitive speech features and depression symptom severity. We observed that (i) speaking rate and articulation rate were significantly associated with daily fluctuations in depression, and (ii) the same features, as well as pitch variability, were significantly associated with longitudinal changes in depression. These results complement growing evidence for the utility of speech as a digital biomarker of depression.

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## 6. References

- [1] S. Dattani, L. Rod s-Guirao, H. Ritchie, and M. Roser, "Mental health," 2023, published online at OurWorldinData.org. [Online]. Available: <https://ourworldindata.org/mental-health>
- [2] K. Mao, Y. Wu, and J. Chen, "A systematic review on automated clinical depression diagnosis," *npj Mental Health Research*, vol. 2, no. 1, 2023. [Online]. Available: <https://doi.org/10.1038/s44184-023-00040-z>
- [3] E. Kato, A. E. Borsky, S. H. Zuvekas, A. Soni, and Q. Ngo-Metzger, "Missed opportunities for depression screening and treatment in the United States," *The Journal of the American Board of Family Medicine*, vol. 31, no. 3, pp. 389–397, 2018. [Online]. Available: <https://doi.org/10.3122/jabfm.2018.03.170406>
- [4] H. Hsin, M. Fromer, B. Peterson, C. Walter, M. Fleck, A. Campbell, P. Varghese, and R. Califf, "Transforming Psychiatry into Data-Driven Medicine with Digital Measurement Tools," *npj Digital Medicine*, vol. 1, no. 1, pp. 1–4, 2018. [Online]. Available: <http://dx.doi.org/10.1038/s41746-018-0046-0>
- [5] A. Coravos, S. Khozin, and K. D. Mandl, "Developing and adopting safe and effective digital biomarkers to improve patient outcomes," *NPJ digital medicine*, vol. 2, no. 1, p. 14, 2019.
- [6] N. Cummins, S. Scherer, J. Krajewski, S. Schnieder, J. Epps, and T. F. Quatieri, "A review of depression and suicide risk assessment using speech analysis," *Speech communication*, vol. 71, pp. 10–49, 2015. [Online]. Available: <https://doi.org/10.1016/j.specom.2015.03.004>
- [7] D. M. Low, K. H. Bentley, and S. S. Ghosh, "Automated assessment of psychiatric disorders using speech: A systematic review," *Laryngoscope investigative otolaryngology*, vol. 5, no. 1, pp. 96–116, 2020. [Online]. Available: <https://doi.org/10.1002/lio2.354>
- [8] J. C. Mundt, P. J. Snyder, M. S. Cannizzaro, K. Chappie, and D. S. Gerals, "Voice acoustic measures of depression severity and treatment response collected via interactive voice response (IVR) technology," *Journal of Neurolinguistics*, vol. 20, no. 1, pp. 50–64, 2007. [Online]. Available: <https://doi.org/10.1016/j.jneuroling.2006.04.001>
- [9] T. F. Quatieri and N. Malyska, "Vocal-source biomarkers for depression: A link to psychomotor activity," in *Interspeech 2012*. Portland, OR, USA: ISCA, 2012, pp. 1059–1062. [Online]. Available: <https://doi.org/10.21437/Interspeech.2012-311>
- [10] N. Cummins, J. Dineley, P. Conde, F. Matcham, S. Siddi, F. Lamers, E. Carr, G. Lavelle, D. Leightley, K. M. White *et al.*, "Multilingual markers of depression in remotely collected speech samples: A preliminary analysis," *Journal of Affective Disorders*, vol. 341, pp. 128–136, 2023. [Online]. Available: <https://doi.org/10.1016/j.jad.2023.08.097>
- [11] J. Gratch, R. Artstein, G. Lucas, G. Stratou, S. Scherer, A. Nazarian, R. Wood, J. Boberg, D. DeVault, S. Marsella, D. Traum, S. Rizzo, and L.-P. Morency, "The Distress Analysis Interview Corpus of Human and Computer Interviews," in *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC'14)*. Reykjavik, Iceland: European Language Resources Association (ELRA), 2014, pp. 3123–3128. [Online]. Available: <https://aclanthology.org/L14-1421/>
- [12] M. Hamilton, "A rating scale for depression," *Journal of Neurology, Neurosurgery, and Psychiatry*, vol. 23, no. 1, p. 56, 1960.
- [13] K. Kroenke, R. L. Spitzer, and J. B. Williams, "The PHQ-9: validity of a brief depression severity measure," *Journal of General Internal Medicine*, vol. 16, no. 9, pp. 606–613, 2001. [Online]. Available: <https://doi.org/10.1046/j.1525-1497.2001.016009606.x>
- [14] Y. Benjamini and Y. Hochberg, "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 57, no. 1, pp. 289–300, 1995. [Online]. Available: <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- [15] L. Hoffman, *Longitudinal Analysis: Modeling Within-Person Fluctuation and Change*. New York, NY, USA: Routledge, 2015.
- [16] Silero Team, "Silero VAD: pre-trained enterprise-grade Voice Activity Detector (VAD), Number Detector and Language Classifier," 2024, GitHub. [Online]. Available: <https://github.com/snakers4/silero-vad>
- [17] E. Harper, S. Majumdar, O. Kuchaiev, L. Jason, Y. Zhang, E. Bakhturina, V. Noroozi, S. Subramanian, K. Nithin, H. Jocelyn, F. Jia, J. Balam, X. Yang, M. Livne, Y. Dong, S. Naren, and B. Ginsburg, "NeMo: a toolkit for Conversational AI and Large Language Models," [Online]. Available: <https://github.com/NVIDIA/NeMo>
- [18] Y. Jadoul, B. Thompson, and B. de Boer, "Introducing parselmouth: A python interface to praat," *Journal of Phonetics*, vol. 71, pp. 1–15, 2018. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0095447017301389>
- [19] N. Cummins, L. L. White, Z. Rahman, C. Lucas, T. Pan, E. Carr, F. Matcham, J. Downs, R. J. Dobson, and J. Dineley, "A methodological framework and exemplar protocol for the collection and analysis of repeated speech samples," *under review JMIR Research Protocols*, 2025.
- [20] O. Murton, R. Hillman, and D. Mehta, "Cepstral Peak Prominence Values for Clinical Voice Evaluation," *American Journal of Speech-Language Pathology*, vol. 29, no. 3, pp. 1596–1607, 2020. [Online]. Available: <https://doi.org/10.1044/2020-AJSLP-20-0000>
- [21] M. Maechler, P. Rousseeuw, C. Croux, V. Todorov, A. Ruckstuhl, M. Salibian-Barrera, T. Verbeke, M. Koller, E. L. T. Conceicao, and M. Anna di Palma, *robustbase: Basic Robust Statistics*, 2024, R package version 0.99-4-1. [Online]. Available: <http://robustbase.r-forge.r-project.org/>
- [22] P. J. Curran and D. J. Bauer, "The Disaggregation of Within-Person and Between-Person Effects in Longitudinal Models of Change," *Annual Review of Psychology*, vol. 62, no. Volume 62, 2011, pp. 583–619, Jan. 2011, publisher: Annual Reviews. [Online]. Available: <https://www.annualreviews.org/content/journals/10.1146/annurev.psych.093008.100356>
- [23] F. L. Huang, W. Wiedermann, and B. Zhang, "Accounting for Heteroskedasticity Resulting from Between-Group Differences in Multilevel Models," *Multivariate Behavioral Research*, vol. 58, no. 3, pp. 637–657, 2022. [Online]. Available: <https://doi.org/10.1080/00273171.2022.2077290>
- [24] J. Pinheiro, D. Bates, and R Core Team, *nlme: Linear and Nonlinear Mixed Effects Models*, 2025, R package version 3.1-167. [Online]. Available: <https://CRAN.R-project.org/package=nlme>
- [25] J. Pustejovsky, *clubSandwich: Cluster-Robust (Sandwich) Variance Estimators with Small-Sample Corrections*, 2024, R package version 0.5.11.9999. [Online]. Available: <http://jepusto.github.io/clubSandwich/>
- [26] J. C. Mundt, A. P. Vogel, D. E. Feltner, and W. R. Lenderking, "Vocal acoustic biomarkers of depression severity and treatment response," *Biological Psychiatry*, vol. 72, no. 7, pp. 580–587, 2012. [Online]. Available: <https://doi.org/10.1016/j.biopsych.2012.03.015>
- [27] A. C. Trevino, T. F. Quatieri, and N. Malyska, "Phonologically-based biomarkers for major depressive disorder," *EURASIP Journal on Advances in Signal Processing*, vol. 2011, pp. 1–18, 2011.
- [28] J. Dineley, E. Carr, F. Matcham, J. Downs, R. J. B. Dobson, T. F. Quatieri, and N. Cummins, "Towards robust paralinguistic assessment for real-world mobile health (mHealth) monitoring: an initial study of reverberation effects on speech," in *Interspeech 2023*. Dublin, Ireland: ISCA, 2023, pp. 2373–2377. [Online]. Available: <https://doi.org/10.21437/Interspeech.2023-947>